



ARDUINO PROJECT

Smart Baby Cradle Controller

An intelligent Arduino-powered system that monitors a baby's crying, diaper condition, and automatically soothes with cradle rocking, ambient light, and music.

Automated • Safe • Intelligent

WHAT IS THE SMART BABY CRADLE?

The Smart Baby Cradle is an Arduino-based automated baby monitoring and soothing system. It detects when a baby is crying using a sound sensor, checks for a wet diaper using a moisture sensor, and responds automatically — rocking the cradle with a servo motor, fading ambient LED light, playing music via a speaker relay, and alerting parents with a buzzer.



Baby Comfort

Automatically rocks and soothes the baby when crying is detected — no parent needed.



Diaper Alert

Moisture sensor detects a wet diaper and notifies parents with a buzzer. The buzzer also alerts when crying is detected by the sound sensor.



Ambient Response

LED light fades gently in and out to create a calm environment for the baby.

COMPONENTS USED

Arduino Uno

Main microcontroller — runs all logic

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Buzzer

Beeps 3× to alert parents of wet diaper

D4

Moisture Sensor

Detects wet diaper (analog input)

A0

Relay Module

Switches music speaker ON/OFF

D9

Sound Sensor (KY-038)

Detects baby crying (digital HIGH)

D2

Servo Motor (SG90)

Rocks cradle: 45° ↔ 135° motion

D6

LED (PWM)

Ambient light — fades in/out smoothly

D3

INPUTS — Sensors



Moisture Sensor

Pin: A0 | Type: Analog

- Reads a value from 0–1023 continuously
- Low value = WET (water conducts electricity)
- Threshold set at 800 — below = diaper wet
- Triggers buzzer to notify parents
- Works like a resistance bridge in water



Sound Sensor (KY-038)

Pin: D2 | Type: Digital

- Reads HIGH (1) when sound is detected
- LOW (0) = quiet — no action taken
- Sensitivity adjusted via potentiometer
- Triggers rocking, LED, music relay, and buzzer
- Uses digitalRead() for fast response

OUTPUTS — Actuators



Servo Motor (Cradle)

D6

- Rocks baby from 45° to 135°
- Smooth movement via smoothMove()
- Returns to 90° center when quiet
- Uses 12ms delay per degree step



LED Light (PWM)

D3

- Fades from 0 to 120 brightness
- Fades back down smoothly
- Stops instantly if crying stops
- analogWrite() controls brightness



Buzzer

D4

- 3 beeps on wet diaper OR crying detected
- HIGH = ON, LOW = OFF
- 300ms ON / 300ms OFF pattern
- Triggered by moisture AND sound sensor



Relay (Music Speaker)

D9

- Turns speaker ON when crying
- Relay switches 5V to speaker
- Turns OFF when baby is quiet
- digitalWrite HIGH = speaker ON

FUNCTIONS — How Each Works

smoothMove(angle)

Moves servo from current angle to target angle one degree at a time with 12ms delay. Prevents jerky motion that could startle the baby.

fadeLED()

Loops brightness from 0→120 then 120→0 using analogWrite(). Exits immediately if crying stops — baby-responsive.

stopAll()

Turns LED off (analogWrite 0), relay OFF, and returns servo to 90°. Called when no crying is detected.

rockCradle()

Calls smoothMove() three times: → 45° (left), → 135° (right), → 90° (center). Creates a gentle rocking motion.

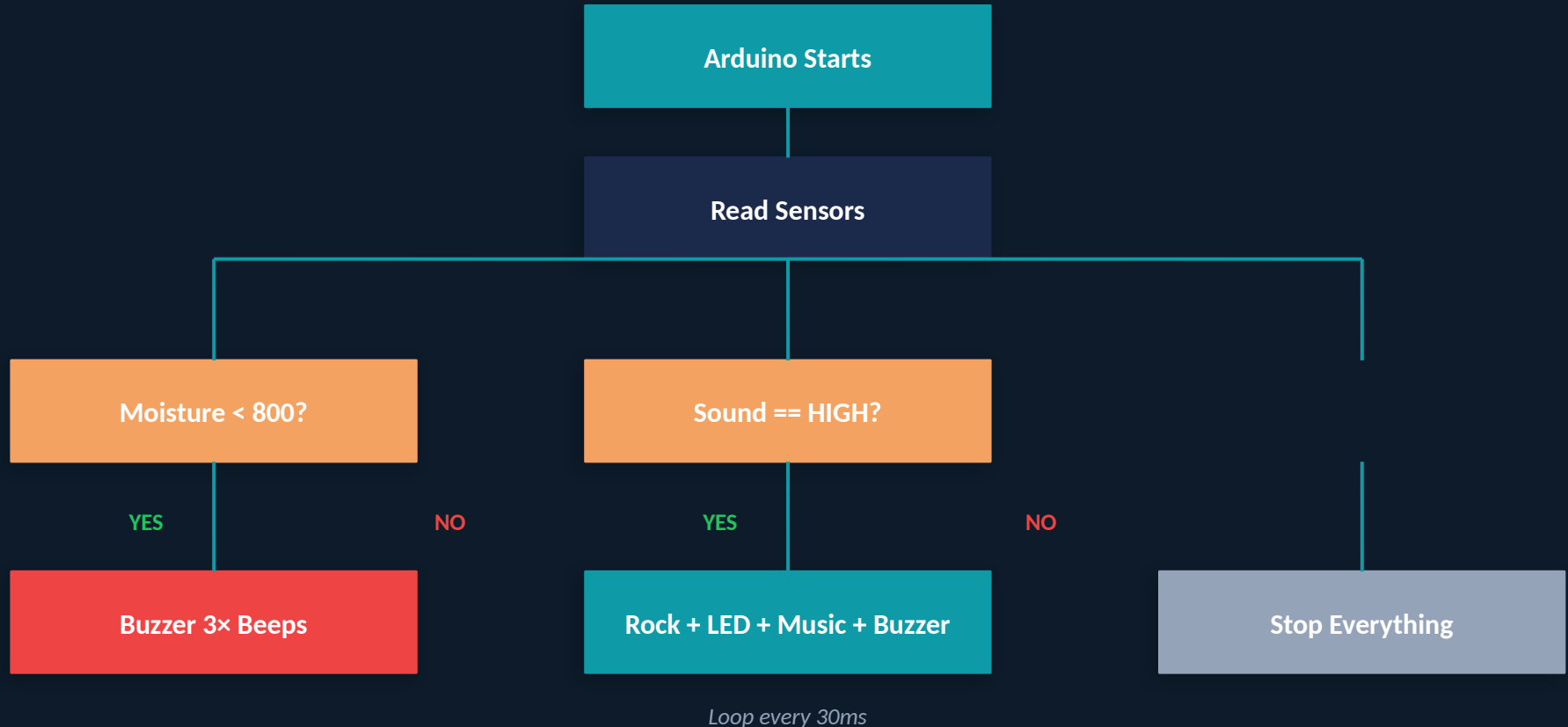
notifyBuzzer()

Runs a loop 3 times: buzzer HIGH 300ms, then LOW 300ms. Produces 3 short alert beeps — triggered by wet diaper (moisture sensor) OR crying detection (sound sensor).

loop()

Main execution loop: reads sensors every 30ms, checks thresholds, and calls appropriate functions based on sensor values.

SYSTEM FLOW DIAGRAM



SUMMARY & PIN REFERENCE

PIN	COMPONENT	TYPE	PURPOSE
A0	Moisture Sensor	Analog IN	Detect wet diaper (< 800 = wet)
D2	Sound Sensor	Digital IN	Detect crying (HIGH = sound)
D3	LED	PWM OUT	Ambient light fade in/out
D4	Buzzer	Digital OUT	Alert beeps for wet diaper AND crying
D6	Servo Motor	PWM OUT	Rock cradle 45°-90°-135°
D9	Relay Module	Digital OUT	Switch music speaker ON/OFF

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Sensors

Moisture + Sound

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Actuators

Servo · LED · Buzzer · Relay

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Functions

Smart automated logic